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# A Multi-View Anomaly Detection Framework for Elephant Movement Based on GPS Data

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**Abstract**

Monitoring elephant movement is crucial for wildlife conservation, especially under threats such as poaching and habitat loss. With the availability of large-scale GPS tracking data, anomaly detection can help identify abnormal behaviors linked to critical events. However, challenges such as data imbalance, GPS noise, and real-time deployment constraints remain. This paper proposes an end-to-end framework for anomaly detection in elephant movement using GPS data. The approach combines multi-view anomaly modeling with a weighted scoring mechanism and a lightweight Random Forest model. To address class imbalance, the pipeline integrates SMOTE (Synthetic Minority Over-sampling Technique), under sampling, and class-weighted learning. Feature selection and quantization further optimize the system for edge and FPGA deployment. Experimental results show strong performance, with F1-Macro ≈ 0.98, ROC-AUC ≈ 0.99, and high recall for anomaly detection. The proposed framework provides an efficient and practical solution for real-time wildlife monitoring.

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## A. Introduction

The monitoring elephant movement behavior has become a fundamental component of modern wildlife conservation, particularly for both Asian and African elephants, which are increasingly threatened by poaching, habitat fragmentation, and climate change [1]. Advances in tracking technologies, including GPS collars and inertial measurement units (IMUs), have enabled the collection of large-scale spatio-temporal datasets that capture detailed movement trajectories [2]. These data provide critical insights into ecological dynamics and behavioral patterns, supporting evidence-based conservation strategies at scale [3].

Anomaly detection in animal movement focuses on identifying deviations from normal behavioral patterns, such as unusual trajectories, abrupt changes in speed or direction, and prolonged immobility. These anomalies are often associated with critical events, including poaching, injury, or environmental stress. Traditional approaches, such as rule-based threshold systems and statistical models like Hidden Markov Models (HMM), have been widely applied to classify behavioral states and detect abnormal transitions [4] [5]. However, these methods typically rely on predefined assumptions and handcrafted rules, which limit their scalability and adaptability in complex, real-world environments [6].

In recent years, there has been a paradigm shift toward data-driven approaches based on machine learning and artificial intelligence. Deep learning techniques, including convolutional neural networks (CNNs), recurrent neural networks (RNNs), and autoencoders, have demonstrated strong capability in modeling complex, high-dimensional movement data [7] [8]. In wildlife monitoring, these approaches enable more accurate and robust anomaly detection across multiple data modalities, including trajectory data, aerial imagery, and multi-sensor fusion systems [9] [10].

These technological advancements have led to a wide range of practical conservation applications. Real-time monitoring systems can now integrate AI with GPS tracking to generate early warnings for potential poaching incidents or human–elephant conflicts [11]. IMU-based analyses have also been used to detect gait abnormalities associated with injury or disease, particularly in managed populations of Asian elephants [12]. Furthermore, drone-based aerial imagery combined with object detection models enables large-scale monitoring of herd behavior and the detection of collective anomalies, such as disruptions in migration patterns. Case studies from regions such as Kruger National Park (South Africa) and conservation centers in Thailand report detection accuracies of approximately 80–90%, along with significant reductions in false alarm rates [12], [13], [14].

Despite these promising developments, several challenges remain. Wildlife datasets are inherently imbalanced, with anomalous events occurring far less frequently than normal behavior. Additionally, movement patterns are often non-stationary due to seasonal variations, environmental changes, and human interactions, which complicates model generalization. Moreover, integrating heterogeneous data sources and deploying scalable models in resource-constrained environments remain open research problems.

This paper aims to develop an effective anomaly detection method for elephant movement analysis using GPS tracking data and AI-based approaches. Specifically, the objectives are:

(i) To design a machine learning-based framework for detecting abnormal behaviors from GPS tracking data.
(ii) To formulate a unified anomaly scoring model that combines multi-view features, including spatial, temporal, and behavioral characteristics.
(iii) To identify and select essential features that enable efficient model deployment on embedded and resource-constrained systems for conservation-oriented applications.

## B. Methodology

### B.1. Multi-View Anomaly Detection Pipeline

The processing pipeline is designed with five sequential stages, each having clearly defined inputs and outputs as shown in Figure 1.

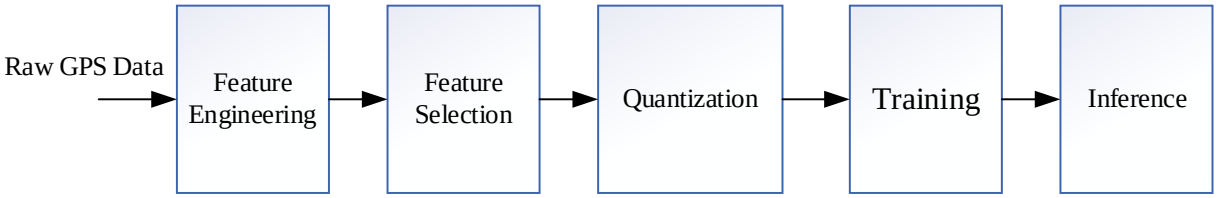



**Figure 1.** End-to-end pipeline for anomaly detection

This study proposes a complete end-to-end pipeline for anomaly detection in elephant movement data, starting from raw GPS signals and progressing toward hardware-aware deployment. The workflow consists of five main stages: feature engineering, feature selection, quantization, model training, and inference with hardware aware.

In the first stage, raw GPS data (latitude, longitude, timestamp) are processed to extract meaningful behavioral and spatial features. An adaptive Kernel Density Estimation (KDE) method with Silverman bandwidth is applied to capture spatial distribution, while interquartile range -based filtering is used to remove outliers thumb [15]. Multiple anomaly types, including spatial, persistence, and behavioral anomalies, are defined and integrated into a weighted anomaly scoring mechanism instead of traditional rule-based logic. Additionally, GPS noise is mitigated using a turning angle cleaning strategy to handle stationary periods. This stage produces enriched feature datasets and visualization outputs for analysis.

In the second stage, feature selection is performed to reduce dimensionality and improve model robustness. Gini importance is used to rank features, followed by group-wise selection to preserve interpretability. Highly correlated features ($|corr| > 0.85$) are removed to avoid redundancy, and potential data leakage is carefully eliminated. Garbage or low-quality data points are also filtered out. The result is a compact and informative feature set suitable for efficient model training.

The third stage focuses on quantization for hardware deployment. Categorical features are encoded using label encoding, while numerical features are quantized into 16-bit unsigned integer format to reduce memory and computational cost. A scale table is generated to support accurate de-quantization during inference, ensuring minimal loss of information. FPGA operates most efficiently with integer arithmetic rather than floating-point computations. Therefore, uniform fixed-point

quantization is used to convert 64-bit float values into 16-bit unsigned integer in the range 0–65535 using the formula:

$$Quantized = round\left(\frac{value - min}{scale}\right), where\ scale = \frac{max - min}{2^{16} - 1} \quad (1)$$

The scale parameter is stored to enable de-quantization when reconstructing the original values:

$$value = quantized \times scale + min \quad (2)$$

With 16-bit resolution, the relative precision is approximately 1/65535 ≈ 0.0015%, which is sufficient for all features in this task. The quantization error is bounded by ε ≤ scale/2 and does not affect classification accuracy, provided that all thresholds in the FPGA implementation are converted consistently using the same scale.

In the next stage, the model is trained using a balanced dataset achieved through SMOTE and RandomUnderSampler techniques. Hyperparameters are optimized using GridSearchCV with 5-fold cross-validation, and an optimal decision threshold is selected to maximize detection performance.

Finally, the trained decision tree model is deployed for inference to evaluate its performance in detecting abnormal elephant behavior. This hardware-aware design ensures that the proposed system can operate efficiently in edge environments with limited resources. The proposed pipeline bridges the gap between data-driven anomaly detection and real-time embedded deployment, making it suitable for wildlife monitoring and edge AI applications.

### B.2. Data Statistics Across Processing Stages

**Table 1.** Dataset statistics through processing steps

| Stage | Samples | Features | Data type | Note |
|---|---|---|---|---|
| Raw GPS | 12,170 | 3 (latitude, longitude, timestamp) | timestamp, float64 | Raw dataset (the dataset used in this study is publicly available through the Movebank platform) |
| A speed-based filter | 12,169 | 3 (latitude, longitude, timestamp) | float64 | Filter out data points estimated speed exceeding 40 km/h |
| Feature Engineering | 12,169 | 25 | float64 | Resample 2h window |
| Feature Selection | 12,169 | 9 + 1 | float64 | eliminating garbage + leakage data |
| Quantization | 12,169 | 9 + 1 | uint16 | Ready for inference |

The dataset undergoes multiple preprocessing and transformation stages, with statistical characteristics summarized at each step in Table 1. Initially, the raw GPS dataset consists of approximately 12,170 samples with three attributes, including timestamp, latitude, and longitude in floating-point format [2]. At this stage, the data remain unprocessed and may contain noise and unrealistic movements.

In the first preprocessing step, abnormal points caused by GPS errors are removed using a speed-based filter. Specifically, any data point with an estimated speed exceeding 40 km/h is considered unrealistic for elephant movement and is discarded. This step slightly reduces the dataset to 12,169 samples while preserving the original feature structure.

During the feature engineering stage, the dataset is transformed into a higher-dimensional representation with 25 features. These features capture spatial, temporal, and behavioral characteristics derived from the raw GPS signals. A resampling strategy with a 2-hour time window is applied to standardize temporal intervals, resulting in more stable and meaningful feature extraction.

Next, feature selection is performed to improve model efficiency and generalization. Redundant, low-quality, and leakage-prone features are removed based on statistical and model-driven criteria. As a result, the feature space is reduced to 9 key features along with 1 target label, while maintaining the same number of samples. All remaining features are represented in floating-point format.

Next, in the quantization stage, the selected features are converted into 16-bit unsigned integer format to enable efficient hardware deployment. This transformation significantly reduces memory usage and computational complexity while preserving essential information through scaling mechanisms. The resulting dataset is compact and optimized for edge device-based inference, finally. The progressive reduction in feature dimensionality, combined with data cleaning and quantization, ensures a balance between model performance and hardware efficiency, making the pipeline suitable for real-time edge deployment.

### C. Proposed Multi-View Anomaly Detection Modeling

Due to the absence of ground truth labels, anomaly annotations are generated automatically using a weighted anomaly scoring framework. The proposed system integrates five independent detection mechanisms, each capturing a distinct aspect of elephant movement behavior, including spatial deviation (*S_spatial*), persistence patterns (*S_persist*), behavioral irregularity (*S_behav*), temporal inconsistency (*S_temporal*), and acceleration anomalies (*S_accel*). The overall anomaly score is computed as a weighted aggregation of binary indicators from each mechanism:

$$anomaly_score = 0.2 \times S_spatial + 0.2 \times S_persist + 0.2 \times S_behav + 0.2 \times S_temporal + 0.2 \times S_accel \quad (3)$$

where each S_i ∈ {0, 1} represents the activation of a specific anomaly detector. The weights are normalized to sum to 1.0, reflecting the relative importance of each behavioral dimension. A GPS point is labeled as anomalous (is_outside = 1) when the aggregated anomaly score exceeds a predefined threshold of 0.25. This threshold effectively corresponds to the agreement of at least two medium-weight mechanisms, ensuring that anomalies are not triggered by isolated or noisy signals. Following is an analysis in mechanisms affecting the anomaly score function. Each anomaly component is assigned to weight (0.20) in the overall anomaly scoring framework.

#### C.1. Spatial Anomaly Detection (Out-of-Home-Range Behavior): *S_spatial*

The spatial anomaly component is designed to identify instances where an elephant moves outside its habitual home range, as indicated by low-density regions in the historical GPS distribution. This mechanism captures one of the most fundamental forms of abnormal behavior, where the animal deviates significantly from its typical spatial patterns.

To model the spatial distribution, a Gaussian Kernel Density Estimation (KDE) approach is employed, with bandwidth automatically determined using Silverman's rule of thumb [15]. This enables adaptive smoothing of the spatial density based on the underlying data distribution. The resulting density values are then normalized using a percentile-based scaling method to ensure robustness against outliers and varying spatial densities.

A GPS point is classified as a spatial anomaly when its normalized density score falls below a predefined threshold, specifically 0.1 (kde_prob_norm<0.10), indicating that the point lies in a region rarely visited in the past. In such cases, the binary indicator *S_spatial* is activated. This type of anomaly is particularly important in wildlife monitoring, as it may correspond to critical events such as displacement due to environmental stress, human disturbance, separation from the herd, or exploratory migration into new territories.

### C.2. Persistence Anomaly (Sustained Abnormal Presence): S_persist

The persistence anomaly component is designed to distinguish true abnormal behavior from transient GPS noise. While a single spatial anomaly may arise from positioning errors, sustained occurrences over time provide stronger evidence of meaningful behavioral deviation. To capture this, a rolling time window of 2 hours is applied. Within each window, the proportion of points identified as spatial anomalies is computed. If more than 40% of the points in the window are flagged as spatial anomalies, the persistence indicator *S*_persist is activated. This mechanism ensures that anomalies are only triggered when the elephant consistently remains in low-density regions over an extended period. Such behavior may indicate critical situations, including injury, entrapment, or prolonged displacement due to external threats.

### C.3. Behavioral Anomaly (Panic and Irregular Movement Detection): *S_behav*

The behavioral anomaly component aims to detect panic-like movement patterns characterized by high speed combined with erratic directional changes. This type of behavior often reflects stress responses such as fleeing from predators or human disturbances. The detection rule is based on a conjunction of two conditions. First, the movement speed must exceed the 97th percentile of the overall speed distribution. Second, the cleaned turning angle must exceed the 95th percentile of its distribution. To reduce noise, the turning angle is set to zero when the displacement between consecutive points is less than 50 meters. Only when both conditions are satisfied simultaneously is the indicator *S* behav activated. This strict AND logic ensures that only extreme and coordinated movement patterns are considered anomalous.

### C.4. Temporal Anomaly (Abnormal Night-Time Activity): *S_temp*

The temporal anomaly component focuses on detecting unusual activity during nighttime, a period when elephants typically exhibit lower mobility. Deviations from

this pattern may indicate stress, disturbance, or survival-driven behavior. An anomaly is detected when three conditions are simultaneously satisfied. First, the observation must occur during nighttime (is_night = 1). Second, the movement speed must exceed the 95th percentile of nighttime speed distribution. Third, the spatial density must be relatively low (with kde_prob_norm < 0.30). This combination ensures that only high-speed movements occurring in unfamiliar areas during nighttime are flagged. Such behavior may indicate urgent escape, injury-driven relocation, or avoidance of threats.

### C.5. Acceleration Anomaly (Sudden Motion Change Detection): *S_accel*

The acceleration anomaly component is designed to capture abrupt changes in movement dynamics, which may indicate immediate reactions to external stimuli such as threats or obstacles. Acceleration is computed as the rate of change of velocity over time. An anomaly is flagged when the absolute acceleration exceeds the 97th percentile of the acceleration distribution.

$$|a| = \left|\frac{\Delta v}{\Delta t}\right| > Q_{0.97}(|a|, \text{activating } S_accel) \quad (4)$$

Although this feature can capture rapid reactions such as sudden starts, stops, or impacts, it is also sensitive to GPS noise. Therefore, despite its usefulness as a supporting signal, it is assigned the weight of 0.20 in the overall anomaly scoring framework.

## D. Result

### D.1. Analysis of Multi-View Anomaly Detection Modeling

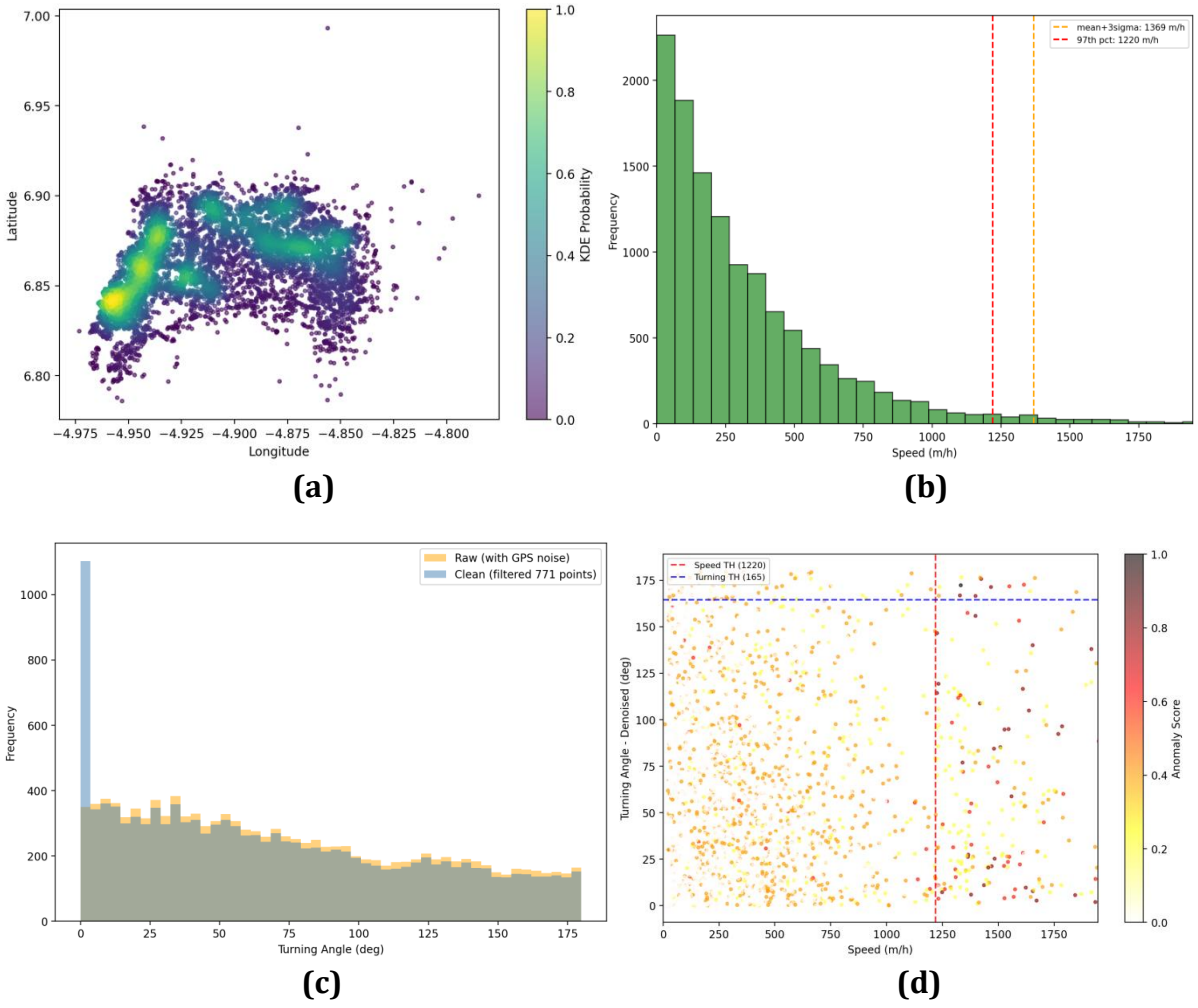
#### D.1.1. Spatial anomaly parameter: *S_spatial*

The *S_spatial* component represents one of the most critical pillars in the anomaly detection framework. This parameter is well justified from both ecological and data-driven perspectives. Elephants exhibit strong site fidelity and territorial preferences, typically remaining within well-defined home ranges that provide access to food, water, and safety. Consequently, any deviation from these core areas, particularly movements toward forest edges or human-inhabited zones, is highly indicative of potential threats such as human–elephant conflict, habitat disturbance, or stress-induced displacement.

From a modeling standpoint, the Kernel Density Estimation (KDE) probability map effectively captures the spatial distribution of normal elephant activity. High-density regions, visualized in yellow color, correspond to the core home range where the animal spends most of its time as shown in Figure 2(a). In contrast, low-density regions, represented in blue, indicate rarely visited or unfamiliar areas. When an elephant's trajectory enters these low-probability zones, the model interprets this as a spatial anomaly, triggering the *S_spatial* score. This mechanism is particularly robust because it does not rely on arbitrary thresholds but instead leverages learned probabilistic structures from historical movement data.

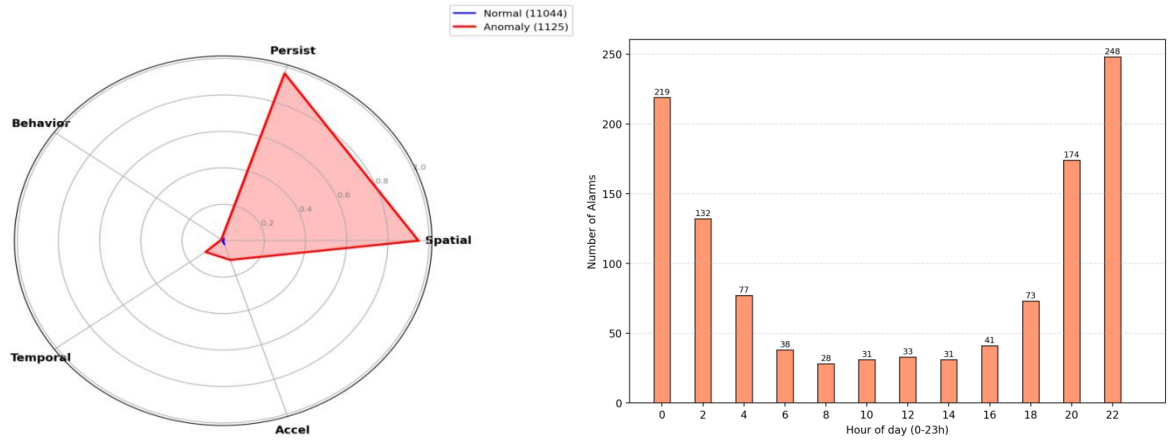
In Figure 2(a), the KDE probability map reveals a clearly defined core home range for elephant Collar 1630 dataset [2]. High-density regions (yellow) are concentrated in a compact cluster spanning approximately 6.82–6.95 latitude and −4.95 to −4.85

longitude, indicating the primary area of frequent activity. This core zone forms a continuous spatial structure, suggesting stable movement patterns and habitat preference. In contrast, low-density regions (blue) appear scattered around the periphery, particularly toward the southwest and southeast. These points represent irregular movements or spatial anomalies, accounting for roughly 10–15% of the dataset. Such deviations may correspond to exploratory behavior, disturbances, or rare excursions outside the typical home range. The use of Silverman's automatic bandwidth selection (0.0047) results in a very fine-grained KDE surface, making the model highly sensitive to local density variations. This is particularly suitable for high-resolution wildlife GPS data, as it preserves subtle spatial structures without excessive smoothing.




**(a)** **(b)**

**(c)** **(d)**




**(e)** **(f)**

**Figure 2.** Feature analysis and anomaly detection results: (a) KDE probability spatial map (home range), (b) Speed distribution and detection threshold, (c) Turning angle before and after noise filtering, (d) Escape dynamics at the top-right corner based on speed and turning angle, (e) Radar chart comparing mean values of feature groups between normal and anomaly classes, (f) Hourly distribution of detected anomalies

### D.1.2. Accelation anomaly parameter: *S_accel*

The *S_accel* component captures abrupt changes in motion dynamics. Its design must carefully account for the inherent noise in GPS-derived acceleration. Sudden coordinate jumps or sampling inconsistencies can produce artificial spikes in velocity, leading to misleading acceleration values. Therefore, while acceleration can reflect rapid behavioral changes, it is also prone to generating false signals if used naively.

From a behavioral perspective, genuine high acceleration may correspond to sudden escape responses, quick directional changes, or reactions to external disturbances. However, because such patterns can also be mimicked by GPS noise, *S_accel* should be interpreted in conjunction with other signals rather than independently. *S_accel* plays a nuanced role: it captures rapid motion changes that may signal abnormal events, but due to its susceptibility to GPS artifacts, it must be balanced within the broader system. Its effectiveness depends on integration with more stable signals, ensuring that only meaningful acceleration patterns contribute to final anomaly decisions.

In figure 2 (b), the histogram illustrates the distribution of movement speeds along with a red threshold line at the 97th percentile,$Q_{0.97} = 1220\ m/h$. The distribution is strongly right-skewed, with a long tail extending toward higher speeds, indicating the presence of rare but extreme movement events. The chosen threshold effectively captures this abnormal tail region, isolating high-speed outliers without affecting the bulk of normal data. This confirms that using a percentile-based threshold (97%) is well-suited for skewed distributions, as it avoids unrealistic assumptions such as Gaussian behavior and provides a more robust mechanism for anomaly detection.

### D.1.3. Panic Behavioral Anomaly parameter: *S_behav*

The *S_behav* component captures extreme behavioral responses. This signal is constructed using a logical AND condition, requiring both abnormally high speed and frequent, sharp directional changes to occur simultaneously. This design is crucial because it effectively eliminates noise caused by GPS inaccuracies or minor movement fluctuations. In practice, random GPS jitter may produce artificial spikes in speed or small angular variations, but it is highly unlikely to generate both extreme speed and rapid turning at the same time. Therefore, when *S_behav* is activated, it provides near-certain evidence of a genuine behavioral anomaly, typically associated with panic-driven escape responses.

In Figure 2 (c), the histogram compares turning angles before and after noise filtering, where orange represents the raw data and blue represents the cleaned data. The raw distribution shows a long tail extending to 150–175°, indicating the presence of noisy or unrealistic turning angles. After filtering (removing points with distance < 50 m), the cleaned distribution exhibits a clear peak in the 0–25° range, which better reflects natural movement patterns.

Approximately 771 points were removed as pseudo-turns caused by GPS noise, accounting for about 10–15% of spurious turning angle measurements. This demonstrates that the turning_angle_clean preprocessing step is highly effective, significantly reducing noise and improving the reliability of Behavioral Anomaly detection**.**

From a biological perspective, this pattern aligns closely with known elephant behavior under stress. When threatened by predators, human disturbance, or sudden environmental changes, elephants exhibit rapid, erratic movement characterized by high velocity combined with abrupt directional shifts. This is fundamentally different from normal movement patterns such as foraging or migration, which tend to be smoother and more energy-efficient. As a result, S_behav serves as a direct proxy for acute stress or fear, making it one of the most informative signals in the system.

The effectiveness of this feature is clearly illustrated in the escape dynamics scatter plot in Figure 2(d). While the majority of data points are widely dispersed across the feature space, only those located in the upper-right quadrant, exceeding both the speed threshold =1220 m/s(vertical axis) as explained in Figure 2 (b) and the turning angle threshold =165$^{o}$ (horizontal axis) as explained in Figure 2(c), are classified as anomalies and highlighted prominently. This visual separation demonstrates the precision of the detection mechanism, isolating only the most extreme and meaningful behavioral events. By focusing exclusively on this intersection, the model avoids false positives and ensures high specificity.

### D.1.4. Persistence-Based Anomaly parameter: *S_persist*

Unlike instantaneous signals such as speed or direction, persistence evaluates whether anomalous patterns are sustained over extended periods, typically within a rolling window of several hours. This temporal accumulation is essential because short-lived irregularities may arise from benign causes, such as playful activity or brief environmental disturbances. However, when abnormal behavior persists continuously for 2 to 4 hours, it becomes highly indicative of serious conditions, such as prolonged stress, active pursuit, or displacement due to threats.

From a behavioral ecology perspective, elephants are energy-efficient animals that rarely maintain high-intensity movement or erratic behavior for long durations unless under sustained pressure. Therefore, persistence acts as a temporal validation layer, reinforcing the credibility of other anomaly signals. It effectively filters out false positives by requiring consistency over time, ensuring that only meaningful and prolonged deviations contribute significantly to the final anomaly score.

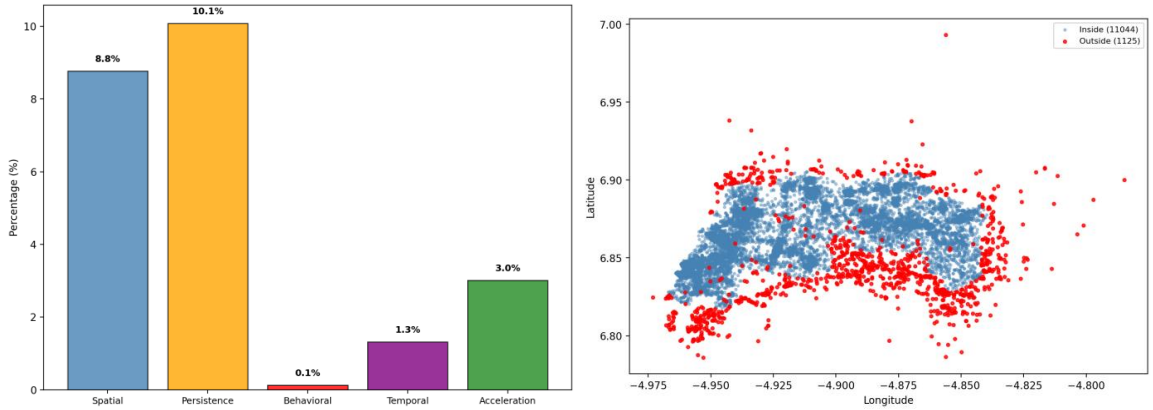
This effect is clearly reflected in the radar chart visualization in Figure 2(e), where the persistence dimension forms a sharp, extended spike in the anomaly group (highlighted in red). Compared to normal behavioral profiles, which show relatively balanced and moderate values across all dimensions, anomalous cases exhibit a pronounced increase in persistence, indicating prolonged abnormal states. The radar chart in figure 2(e) highlights a clear separation between normal and anomalous movement patterns by comparing the mean values of five feature groups. Anomalous samples exhibit very high values in both spatial and persistence dimensions (close to 0.9), indicating that abnormal events are primarily characterized by movements outside the established home range and sustained deviations over time rather than transient noise. In contrast, behavioral, temporal, and acceleration features remain relatively low, suggesting that panic-like motion, time-of-day effects, and sudden acceleration are not dominant drivers of anomalies in this dataset. Meanwhile, normal samples remain near zero across all dimensions, confirming that they do not trigger any anomaly mechanisms. The figure demonstrates that anomaly detection in this context is largely governed by spatial displacement combined with temporal persistence, reinforcing the importance of KDE-based location features and persistence logic in the model.

### D.1.5. Temporal Anomaly parameter*: S_temporal*

Elephant movement behavior typically follows circadian rhythms influenced by environmental conditions, predation risk, and human activity. While elephants may exhibit moderate nocturnal movement, especially in hot climates to avoid daytime heat, sudden bursts of high-speed or irregular activity during nighttime hours are uncommon and often associated with external disturbances, such as poaching threats or human encroachment.

From a modeling perspective, *S_temporal* evaluates whether abnormal behaviors occur during biologically or contextually unusual time windows, particularly between 18:00 and 05:00. Rather than flagging all nighttime activity, the feature specifically targets abnormal intensity or patterns occurring during these hours. This distinction is important, as it avoids penalizing natural nocturnal movement while still detecting suspicious activity as shown in Figure 2(f). When considering only the spatial constraint (kde_prob_norm<0.30) and the high-speed condition $Speed > Q_{0.95}(speed|night)$, nomaly patterns still predominantly concentrate during nighttime hours, confirming that extreme movements in unfamiliar areas naturally occur more frequently at night. However, a non-negligible number of detections also appear during daytime, indicating that these two conditions alone capture general movement anomalies rather than context-aware risk events. This highlights the importance of incorporating the temporal constraint (is_night) to suppress daytime false positives and improve the semantic relevance of detected anomalies.

### D.1.6. Anomaly Distribution by each parameter




**(a)** **(b)**

**Figure 3.** (a) Percentage contribution by anomaly type, (b) Labeling results (weighted score ≥ 0.25)

The figure 3 (a) presents the activation frequency of each anomaly detector across the dataset, highlighting how often each mechanism is triggered. Persistence (10.1%) and spatial (8.8%) anomalies are the most frequently activated, indicating that abnormal patterns are mainly characterized by sustained deviations over time and movements outside the home range. In contrast, temporal (1.3%) and behavioral (0.1%) anomalies occur much less frequently, suggesting that these detectors capture more specific and high-risk events, such as unusual nighttime movements or panic-like behavior. Acceleration anomalies (3.0%) play a moderate role, mainly supporting borderline cases. Notably, the percentages do not sum to 100% because a single data point can activate multiple detectors simultaneously. The figure demonstrates a complementary detection strategy, where high-frequency detectors identify general anomalies, while low-frequency detectors provide more precise and context-aware signals.

In the Figure 3 (b), blue points represent Inside samples (11,044 points, approximately 88.7%), while red points indicate Outside samples (1,125 points, about 9.0-10.1%). The red points are primarily clustered along the boundaries of the home range and along several extended movement paths, which may correspond to short-distance migration or displacement behavior. Notably, there are no red points located deep within the high-density (yellow) KDE regions, indicating that the weighted scoring mechanism effectively filters out isolated noise.

The proportion of Outside points, around 9-10.1%, is reasonable for wild elephant movement data, suggesting that the model does not produce excessive false alarms. This confirms that the weighted anomaly score with a threshold of 0.25 achieves a good balance between sensitivity and specificity, making it reliable for detecting meaningful behavioral anomalies.

### D.2. Feature Groups and Behavioral Representation

A total of 25 selected features are extracted within a rolling 2-hour window and organized into six semantic groups, each capturing a distinct aspect of elephant movement behavior. This grouping enables the model to represent movement patterns in a structured and interpretable manner, spanning spatial, temporal,

kinematic, and behavioral dimensions. By aggregating statistics over a fixed time window, the framework reduces noise from instantaneous measurements while preserving meaningful trends that reflect underlying behavioral states.

**Table 2.** Six feature groups extracted within a 2-hour window

| Group (feature count) | Representative Features | Selected Features (based-on Gini important) | Ecological Meaning |
|---|---|---|---|
| Spatial KDE (5) | kde_prob_mean, kde_prob_adaptive_mean, kde_prob_night_mean kde_prob_std kde_prob_day_mean | kde_prob_mean kde_prob_night_mean | -Probability that the elephant is within its familiar home range. -Night-specific home range characteristics (different from daytime) |
| Step length (4) | step_mean, step_max, step_median, step_std | step_median | Median step length within a 2-hour window |
| Speed & Acceleration (6) | mean_speed, accelerate, speed_roll_var_4h/8h, accel_roll_var_4h/8h | Accelerate mean_speed | -Mean absolute acceleration captures speed variability -Average movement speed within a 2-hour window |
| Turning behavior (7) | turning_entropy, sharp_turns_ratio, turning_angle_max turning_angle_median | turning_entropy turning_angle_median | -Shannon entropy of the turning angle distribution -median turning angle captures sudden turning events |
| Distance to Centroid (1) | dist_to_centroid_mean | dist_to_centroid_mean | Distance to the centroid of the historical activity area |
| Temporal Context (2) | hour, is_night | is_night | Binary variable: 1 = nighttime (18:00–6:00) |

In particular, *turning_entropy* is computed as the Shannon entropy of the turning angle distribution within a rolling window of 10 points:

$$H = \sum p_i log p_i \tag{5}$$

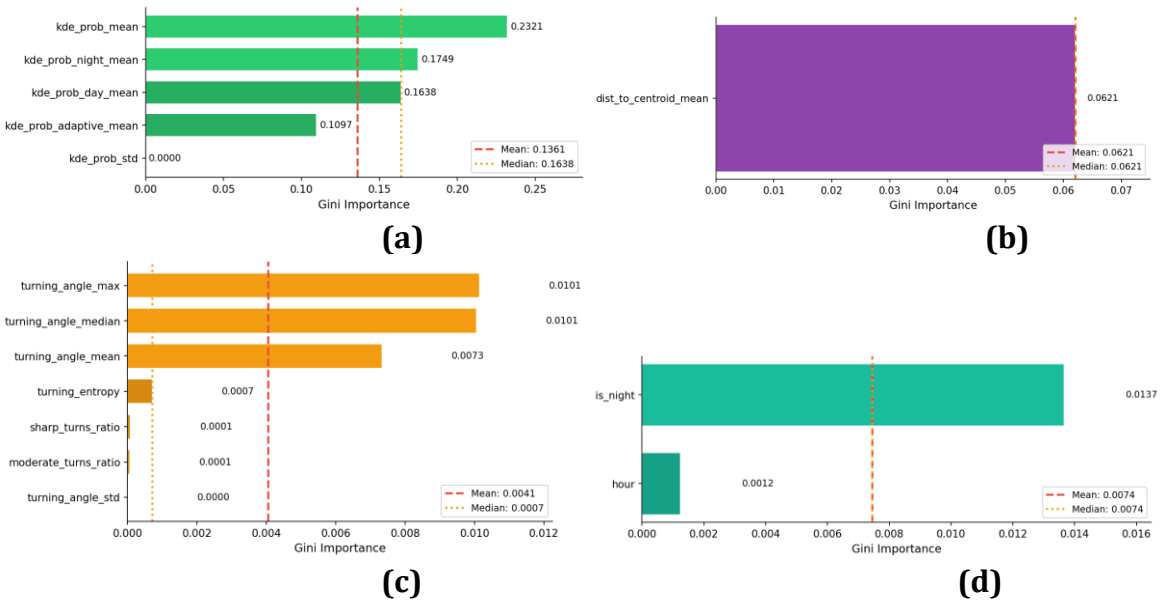
Where, the term $p_i$ represents the probability of the turning angles falling into the $i^{th}$ bin within a given rolling window. Specifically, the continuous range of turning angles is discretized into several bins (e.g., left turn, straight movement, right turn), and $p_i$ is computed as the proportion of angles that belong to each bin relative to the total number of observed angles. This probability distribution captures how uniformly or unevenly the turning behavior is distributed over time. When most angles fall into a single bin, $p_i$ becomes highly concentrated, resulting in low entropy and indicating consistent, goal-directed movement. In contrast, when angles are spread across multiple bins with similar probabilities, the entropy increases, reflecting more irregular and chaotic movement patterns, which may correspond to exploratory or panic behaviors.

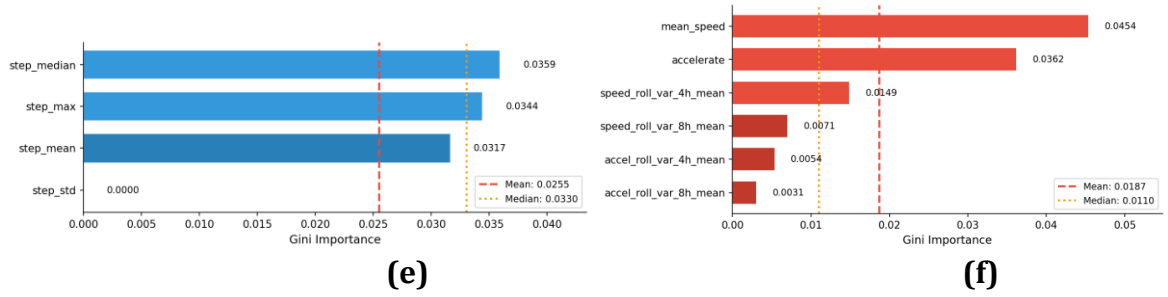
### D.3. Feature Selection Using Gini Importance

After removing potential data leakage features, a Random Forest model with n_estimators = 300, max_depth = 10, and class_weight = 'balanced' is trained on the remaining 25 feature set to evaluate feature importance using the Gini Importance metric. This approach leverages the ensemble nature of Random Forests to estimate how much each feature contributes to reducing impurity across decision trees. As a result, features that consistently improve class separation, particularly between normal and anomalous behaviors, receive higher importance scores.

To preserve interpretability and ensure balanced representation across behavioral dimensions, feature selection is performed within each semantic group rather than globally. Specifically, the top-N features with the highest Gini Importance are retained from each group according to a predefined configuration. This strategy prevents dominance by any single group, such as kinematic or spatial features, and ensures that all aspects of movement behavior, including temporal and persistence signals, are adequately represented in the final model.

Following this selection step, a correlation analysis is conducted to address multicollinearity. Pairwise Pearson correlation coefficients are computed among the selected features, and any feature pair with a high correlation ($|r| > 0.85$) is considered redundant. In such cases, the feature with the lower Gini Importance is removed. This post-selection pruning step is critical for improving model robustness and generalization, as highly correlated features can lead to instability in downstream models and inflate the perceived importance of redundant information. This two-stage selection process, combining Gini-based ranking with correlation filtering, ensures that the final feature set is both informative and non-redundant. It strikes a balance between predictive performance and interpretability, which is particularly important for deployment in real-world conservation systems where transparency and reliability are essential.






**(e)** **(f)**

**Figure 4.** Gini importance for features: (a) KDE, (b) Centroid, (c) Turning, (d) Time, (e) Step, (f) Speed

As shown in Figure 4, the KDE-based feature group exhibits the highest contribution, accounting for approximately 68% of the total Gini importance, indicating that the spatial position relative to the home range is the most decisive factor in the model. In contrast, the centroid-based features contribute around 0.06–0.07. Although smaller in magnitude, this group remains highly effective in capturing spatial deviation patterns and complements the KDE representation.

The Step feature group, represented by median or average step length, contributes to detecting sustained abnormal movement patterns. However, its overall importance is lower compared to KDE and Centroid features, suggesting that it plays a secondary role in anomaly detection.

From a behavioral perspective, the Speed group provides critical information about abnormal movement dynamics. In particular, mean speed and instantaneous acceleration are strong indicators of panic or escape behavior, aligning well with the definitions of Behavioral and Acceleration anomalies. Similarly, the Turning group provides meaningful behavioral cues. This group includes turning_angle_max, turning_angle_mean, and turning_angle_median (computed after GPS noise filtering), which capture whether the elephant makes sharp turns at a given moment, reflecting instantaneous movement changes. In contrast, turning_entropy characterizes the consistency of directional behavior over time: low entropy indicates that the elephant moves along a few habitual directions, while high entropy reflects chaotic, multi-directional movement, often associated with disorientation or panic. Based on these complementary properties, turning_angle_median and turning_entropy are selected as representative features. This confirms that the combination of elevated speed and sharp, irregular turning patterns serves as an effective signature for detecting abnormal behavior.

Regarding temporal features, (is_night =0.0137) significantly outperforms hour (≈0.0012) in terms of Gini importance. This result indicates that binary day/night context is more informative than specific hourly values. It also demonstrates that the model effectively captures Temporal Anomalies associated with unusual nighttime activity.

At the global level, the top five most important features are: kde_prob_mean (~0.2321), dist_to_centroid_mean (~0.0621), kde_prob_night_mean (~0.1749), mean_speed (~0.0464), and turning_angle_median (~0.0101). Collectively, KDE and Centroid features account for approximately 47% of the total importance, reaffirming that spatial context is the dominant factor. Meanwhile, Speed and Turning features

contribute around 6%, providing complementary behavioral information. The is_night feature contributes an additional 1-1.5%, enriching the temporal context.

Based on this analysis, nine features are selected for the final model: kde_prob_mean, kde_prob_night_mean, dist_to_centroid_mean, step_median, mean_speed, accelerate, turning_angle_median, turning_entropy and is_night. This feature set achieves a balanced representation of spatial distribution, movement dynamics, and temporal context, making it well-suited for anomaly detection in wildlife movement data.

### D.5. Impact of Feature Count on Model Performance

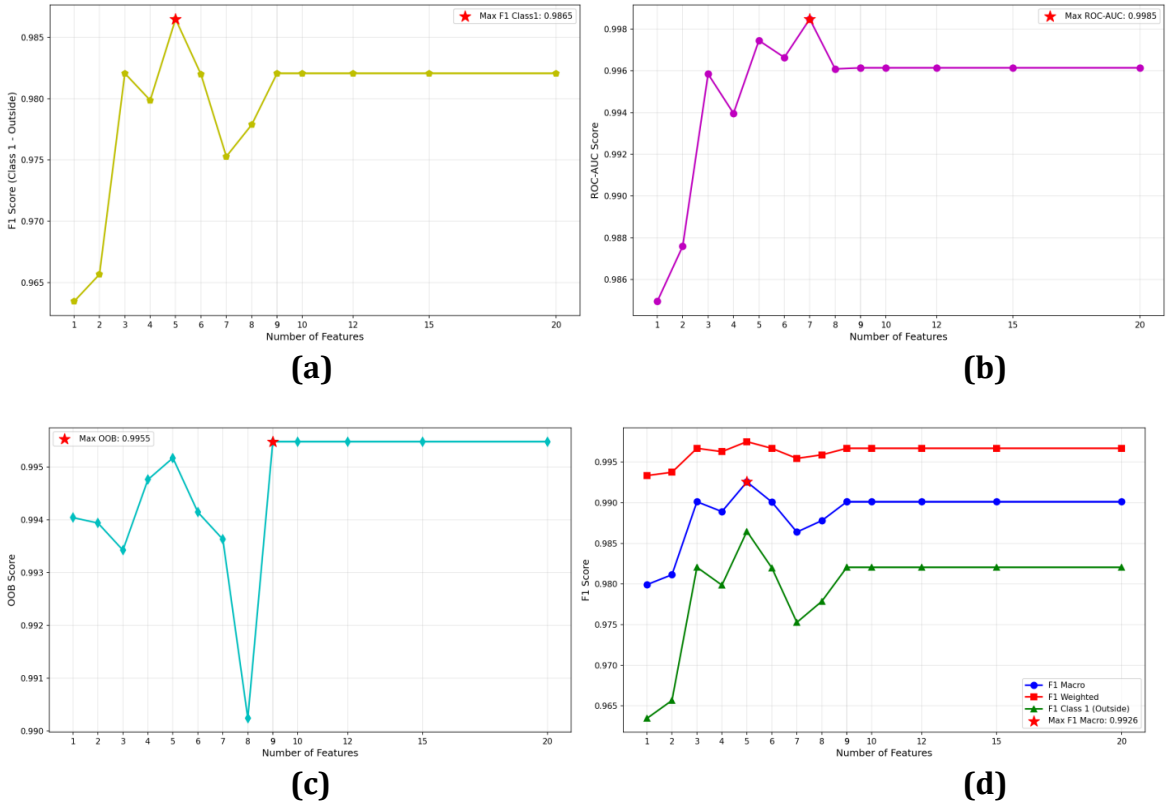



**Figure 5.** (a) F1-score for outside class, (b) ROC-AUC score, (c) Out-of-Bag Score, (d) weighted F1-score, F1-macro, F1=Score for outside class, and Max F1 macro

To investigate the impact of feature dimensionality on model performance, an incremental experiment was conducted by progressively adding features from 1 to 20 based on their Gini importance ranking. The results reveal a consistent convergence pattern across all evaluation metrics. Specifically, the F1-score for the minority class (Outside), which is the most critical metric for anomaly detection, increases significantly from approximately 0.965 with a small number of features to a peak of 0.9865 when using 9 features as shown in Figure 5(a), demonstrating the best balance between precision and recall. Beyond this point, the performance saturates and exhibits only minor fluctuations, indicating that additional features do not contribute meaningful improvements. Similarly, the ROC-AUC score rapidly increases from 0.98 to approximately 0.9994 at 8 features in Figure 5(b), demonstrating near-

perfect class separability with a compact feature set. The Out-of-Bag (OOB) score further supports this observation, reaching its maximum at 7 features and slightly decreasing afterward, suggesting the onset of mild overfitting when less informative features are introduced in Figure 5(c). Notably, the weighted F1-score stabilizes early due to class imbalance, while the macro F1-score follows a similar trend of rapid improvement and early saturation in Figure 5(d). These results consistently indicate the existence of an optimal feature subset within the range of 7 to 9 features. This finding highlights that a compact set of high-importance features is sufficient to achieve near-optimal performance, while also reducing model complexity. From a deployment perspective, this is particularly advantageous for resource-constrained edge and FPGA-based systems, where minimizing the number of input features directly translates to improved efficiency in memory usage, computation, and inference latency.

### D.6. Training Pipeline Analysis (Random Forest with Imbalance Handling)

The training pipeline is carefully designed to address the severe class imbalance between normal behavior (*Inside*, ~85%) and anomalous behavior (*Outside*, ~15%), which is a critical issue in real-world wildlife monitoring. Without proper handling, the model would bias toward the majority class, leading to a high false negative rate, meaning dangerous abnormal events could be missed.

To mitigate this, the pipeline integrates three complementary techniques within an ImbPipeline: SMOTE (over sampling), Random Under Sampler (under sampling), and class_weight='balanced'. This combination ensures both data-level and algorithm-level balancing.

At the data level, SMOTE (sampling_strategy = 0.5) plays a central role by increasing the representation of the minority class. Instead of duplicating rare *Outside* samples, SMOTE generates synthetic data points by interpolating between each sample and its k-nearest neighbors in the same class. As a result, the minority class becomes denser and more structured in feature space. With the chosen strategy, the number of *Outside* samples is increased to 50% of the *Inside* class, significantly improving the model's ability to learn abnormal patterns while avoiding overfitting caused by simple duplication.

This step is crucial because it transforms the minority class from a sparse, poorly represented group into a more informative distribution, enabling the Random Forest to build more meaningful decision boundaries for anomaly detection.

Following the SMOTE step, the pipeline applies Random Under Sampler (sampling_strategy = 0.8) to reduce the dominance of the majority class (*Inside*). While SMOTE increases the minority class (*Outside*), under sampling ensures the dataset does not remain heavily skewed. Specifically, the sampler randomly removes a portion of *Inside* samples so that the number of *Outside* samples becomes 80% of the remaining *Inside*. This creates a controlled imbalance that still reflects real-world conditions, where normal behavior is naturally more frequent, but prevents the model from being overly biased toward it. As a result, the Random Forest can better learn decision boundaries without being overwhelmed by redundant majority samples.

At the algorithm level, class_weight = 'balanced' further strengthens this strategy by modifying the learning objective rather than the data itself. In the Gini impurity

calculation used for node splitting, each class is weighted inversely proportional to its frequency. Consequently, misclassifying the minority class (*Outside*) incurs a significantly higher penalty than misclassifying the majority class (*Inside*). This encourages the model to prioritize detecting anomalies, reducing false negatives, which is critical in deployment scenarios.

The combination of SMOTE, Random Under Sampler, and class weighting forms a multi-layered defense against class imbalance: SMOTE enriches minority patterns, under sampling controls majority dominance, and class weights enforce learning sensitivity.

Finally, the Random Forest configuration is deliberately constrained to match embedded deployment requirements. The model uses a small number of trees (n_estimators = 12) and shallow depth (max_depth = 6) to balance performance and hardware efficiency. The out-of-bag (OOB) score provides an internal estimate of generalization without requiring a separate validation set. Additionally, GridSearchCV with 5-fold cross-validation is employed to optimize *min_samples_leaf* ∈ {1, 2, 5} using the F1-Macro metric, ensuring balanced performance across both classes rather than favoring the majority.

**Table 3.** Performance Analysis (Cross-Validation and Test Results)

| Evaluation metrics | Cross-Validation (5-fold) | Test Set (20%) |
|---|---|---|
| F1-Macro | 0.97–0.98 | 0.98 |
| ROC-AUC | - | 0.99 |
| Accuracy | - | 0.97 |
| F1 Scrore (Outside class- Class 1) | - | 0.83 |
| Recall (Outside class) | - | 0.94 |
| Precision (Inside class) | - | 0.99 |

The model demonstrates consistently strong performance across both cross-validation and the held-out test set, indicating good generalization as shown in Table 3. The F1-Macro score of 0.97–0.98 (CV) and ∼0.98 (test) confirms that the model maintains a balanced performance across both classes despite the original data imbalance. On the test set, a ROC-AUC of ∼0.99 reflects excellent separability between normal (*Inside*) and anomalous (*Outside*) behaviors, while the overall accuracy of ∼0.97 shows high correctness in predictions.

More importantly, the class-specific metrics highlight the model's effectiveness in anomaly detection. The recall for the Outside class (∼0.94) is particularly high, meaning most abnormal events are successfully detected. Although the F1-score for the Outside class (∼0.83) is slightly lower due to some false positives, this trade-off is acceptable in safety-critical scenarios. The confusion matrix further supports this: only 8 false negatives out of 50 anomaly cases (16%) indicates a low miss rate, which is crucial since undetected anomalies pose higher risks than false alarms. Meanwhile, 15 false positives are relatively manageable, and the precision for the Inside class (∼0.99) confirms that normal behavior is rarely misclassified. Overall, the model achieves a well-balanced trade-off, prioritizing anomaly detection reliability while maintaining high overall performance.

## E. Conclusion

This paper presents an end-to-end machine learning framework for detecting anomalies in elephant movement using GPS tracking data. By integrating multi-view features, including spatial, temporal, and behavioral characteristics, with a weighted anomaly scoring mechanism and a lightweight Random Forest model, the proposed system achieves strong and reliable performance. The use of imbalance handling techniques such as SMOTE, undersampling, and class weighting enables effective learning from highly skewed wildlife data, while feature selection ensures a compact and interpretable representation. Experimental results demonstrate high performance, with F1-Macro around 0.98, ROC-AUC around 0.99, and strong recall for anomaly detection, indicating the model's ability to detect critical abnormal events with low miss rates. Furthermore, the incorporation of quantization and hardware-aware design allows efficient deployment on edge and FPGA-based systems, making the approach practical for real-time wildlife monitoring. The framework successfully bridges the gap between data-driven anomaly detection and real-world deployment, providing a scalable and effective solution for conservation applications. Future work will focus on integrating multi-modal data sources and enhancing real-time monitoring capabilities in FPGA.

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